Blueprint A: Constacyclic codes over M4(F2[u]/<u^k>) and quantum codes over F4

Nov 12, 2026 · @Dr. Vipin

Overview

Working title: Hermitian dual-containing constacyclic codes over M4(F2[u]/⟨u^k⟩) and quantum codes over F4.

Core claim: replacing cyclic by λ-constacyclic codes over R = M4(F2[u]/⟨u^k⟩) opens new code lengths and new Hermitian dual-containing codes, which give quantum codes [[N, K, D]]_4 that cyclic codes cannot reach.

Gap this fills: the matrix-ring paper (arXiv 2602.18255) studies only cyclic codes over this ring and stops at classical F16 codes. The quantum paper (Quantum Inf. Process. 25:207) does the CSS construction only for M3(Fp[u]/⟨u^k⟩). The combination M4 + constacyclic + quantum is not covered in either introduction. Both papers' full texts and the literature still need a novelty check.

Draft abstract: Let R = M4(F2[u]/⟨u^k⟩), which is isomorphic to a local ring built on F16 with two nilpotent generators u and v (u^k = 0, v^4 = 0). We study λ-constacyclic codes over R for units λ with λ^5 = 1 in the residue field F16 and for units with a nilpotent part. We describe their structure and cardinality, derive the Euclidean and Hermitian duals, and give a necessary and sufficient condition for Hermitian dual-containment. Through a Gray map to F16 and the Hermitian construction we obtain quantum codes over F4. We list examples with small quantum Singleton defect and compare them with known tables.

Preliminaries and choice of λ

The ring is R = M4(F2[u]/⟨u^k⟩) with |R| = 2^(16k) = 16^(4k). Per the source paper it is isomorphic to A[u]/⟨u^k⟩, where A = F16 + vF16 + v^2 F16 + v^3 F16 and v^4 = 0. It is local with residue field F16 and maximal ideal ⟨u, v⟩.

Units. An element is a unit exactly when its image in F16 is nonzero. So every unit is a0 + (nilpotent part) with a0 in F16*.

Choices of λ.

Family	λ	Why it matters
Scalar	λ in F16* with λ^5 = 1, λ ≠ 1 (exactly four such elements, each of order 5)	Hermitian duality over F16/F4 needs λ^(q+1) = 1 with q = 4, so λ^5 = 1. The dual stays λ-constacyclic.
Scalar, trivial	λ = 1	Baseline: cyclic case from the source paper.
Nilpotent twist	λ = λ0 + u^j (or + v^j), with λ0^5 = 1	Gives a genuinely new ring structure. The dual-containing condition needs a conjugation-compatible check.

Hermitian conjugation on R. Extend a → a^4 on F16 and fix u and v. Open check: this is a ring automorphism only if elements of F16 commute with v in the source paper's isomorphism. The source abstract says u and v commute, but says nothing on F16 versus v, so confirm this in the full text before relying on it.

Length. Take n odd, so x^n - λ is separable over the residue field and lifts by Hensel's lemma. This keeps the CRT decomposition clean.

Structure of λ-constacyclic codes

Goal theorem: classify the ideals of R[x]/⟨x^n - λ⟩ and give |C| in closed form.

Step 1: decomposition. For scalar λ and n odd, factor x^n - λ = f_1 ⋯ f_s over F16 into monic irreducibles. By Hensel lifting and CRT:

```
R[x]/\langle x^n-\lambda\rangle \;\cong\; \bigoplus_{i=1}^{s}
K_i[u,v]/\langle u^k, v^4\rangle, \qquad K_i = \mathbb{F}_{16}[x]/\langle
f_i\rangle
```

So a code is a direct sum C = ⊕ C_i of ideals C_i of the local rings S_i = K_i[u,v]/⟨u^k, v^4⟩.

Step 2: ideals of S_i. S_i is a non-chain Frobenius local ring (a complete intersection), so ideals are not just powers of one generator. Plan: first treat the monomial ideals generated by u^a v^b, which already give a rich family and an explicit count. Then extend to principal ideals and, if feasible, to all ideals via a Gröbner basis argument on K_i[u,v].

Step 3: cardinality. For a monomial ideal I with staircase shape, |I| = |K_i|^(number of monomials u^a v^b in I). Summing over components gives |C| = product of |K_i|^(m_i). State this as a theorem.

Step 4: nilpotent λ. For λ = λ0 + u^j, rewrite x^n - λ as a perturbation of x^n - λ0 and show it still factors by Hensel lifting. Check that the CRT decomposition survives; this is the main new technical point versus the cyclic paper.

Deliverable: Theorem (structure) + Theorem (cardinality) + a worked example for n = 5, k = 2.

Duals and the dual-containing criterion

Key fact: if C is λ-constacyclic, its Euclidean dual is λ^(-1)-constacyclic and its Hermitian dual is λ^(-4)-constacyclic. With λ^5 = 1 we get λ^(-4) = λ, so the Hermitian dual is again λ-constacyclic. This is exactly why λ^5 = 1 is the right scalar family.

Component map. Under the CRT decomposition, the Hermitian dual sends component i to component i′, where i′ is defined by the map on roots j -> -4j (shifted by the λ-coset). Two cases:

1. **Paired components** (i ≠ i′): the condition becomes a containment between the conjugate annihilator of I_(i′) and I_i.
2. **Self-paired components** (i = i′): the condition is I_i contains conj(Ann(I_i)), a self-orthogonality-type condition on one ideal.

Target theorem:

```
C^{\perp_H} \subseteq C \iff \overline{\mathrm{Ann}(C_{i'})} \subseteq C_i \
\text{for every } i
```

For monomial ideals this becomes a purely combinatorial statement on the exponent pairs (a, b), which is what makes a computer search fast.

Why constacyclic helps. The number of self-paired components depends on n and λ. Changing λ changes which cosets are self-paired, so lengths where the cyclic case has too many self-paired cosets (blocking dual-containment) can become feasible.

Proof plan. (a) Prove the dual formula on one component. (b) Prove the annihilator/conjugation compatibility in the Frobenius ring S_i. (c) Assemble via CRT. (d) Check the nilpotent-λ case separately.

Gray map and image over F16

Goal: a map Φ from R^n to F16^(4kn) that sends a λ-constacyclic code to an F16-linear code and keeps Hermitian orthogonality.

Sizes. |R| = 16^(4k), so each ring element maps to 4k symbols of F16. A code of length n over R gives a code of length N′ = 4kn over F16.

Tasks.

1. Reuse the Bachoc-type map and the Gray map from the source paper (the paper's abstract names both). Read their exact definitions in the full text before reusing them.
2. Define a Hermitian-type inner product on R^n and prove Φ carries it to the standard Hermitian product on F16^(4kn). This is the main proof here. If Φ is only an isometry for Euclidean products, a twisted version is needed.
3. Show Φ(C) is a quasi-twisted code of index 4k (for scalar λ of order 5 this is expected, to be verified), which gives a structured generator matrix for fast distance computation.
4. Show Φ(C)^(perp_H) = Φ(C^(perp_H)), so dual-containment transfers.

Distance. Record d_H(Φ(C)). Give a lower bound from the torsion codes (the layers cut out by powers of u and v) as a conjecture to test numerically, not a claim.

Quantum code construction over F4

Hermitian construction. If D is an [N, K, d]_16 code with D^(perp_H) ⊆ D, there is a quantum code [[N, 2K - N, ⩾ d]]_4.

Apply it to D = Φ(C) with N = 4kn:

```
[[\,4kn,\; 2K-4kn,\; D\,]]_4, \qquad K=\dim_{\mathbb{F}_{16}}\Phi(C),\ D \ge d_H(\Phi(C))
```

Quantum Singleton bound for these parameters: K_q + 2D_q <= N + 2, so the defect is delta = N + 2 - K_q - 2D_q. Report delta for every code; delta = 0 is MDS, small delta is the headline.

Parameter formulas to prove.

1. K in terms of the ideal data: K = sum over components of deg(f_i) times (number of monomials in C_i).

2. 2K - N > 0 is equivalent to a lower bound on the total monomial count; state it as a feasibility condition.
3. D >= min distance of Phi(C); also give the pure-code statement when D equals d_H(Phi(C)).

Where better codes can come from. (i) New lengths N = 4kn reachable only for λ ≠ 1. (ii) Same N but different K, from the extra self-paired components. (iii) Codes whose Gray images have good distance because of the u-v layer structure.

Computational search plan

Tools. Magma (best for distance and the Grassl comparison), or GAP with GUAVA, or Python with `galois` for F16 arithmetic. Brouwer-Zimmermann for minimum distance.

Parameter grid.

Parameter	Range	Reason
k	2, 3, 4	Small nilpotency index keeps N = 4kn manageable.
n	odd, up to about 25	Keeps N under about 400 for k up to 4.
λ	1, the four order-5 scalars, and λ0 + u^(k-1)	Cyclic baseline versus new families.
Ideals	monomial ideals u^a v^b first	Explicit counts and fast feasibility test.

Algorithm.

1. Compute the constacyclotomic cosets of x^n - λ over F16 and the pairing i -> i′.
2. Enumerate monomial-ideal choices per component that satisfy the dual-containing condition (pure combinatorics, very fast).
3. For each survivor, compute K, build Φ(C) and its minimum distance (use the quasi-twisted generator matrix).
4. Record [[N, 2K - N, d]]_4 and the defect.

Comparison. Check each result against the online code tables (Grassl) for q = 4 quantum codes. Mark a code *new* only if its parameters are not dominated by a table entry. Also keep codes that match the table or are one step off, since they show the family is competitive.

Output table. Columns: k, n, λ, ideal data, N, K_q, D, defect, status (new / matches / dominated).

Paper outline, work order and risks

Outline.

1. Introduction and motivation (cyclic, matrix rings, quantum codes).
2. Preliminaries: ring R, units, conjugation.
3. Constacyclic codes over R: structure and cardinality.
4. Euclidean and Hermitian duals; dual-containing criterion.
5. Gray map and the F16 image.
6. Quantum codes over F4: parameters and bounds.
7. Examples and comparison table.
8. Conclusion and open problems (determinant condition for space-time use).

Work order.

1. Read both full texts; confirm commutation and the exact Gray and Bachoc maps.
2. Prove the structure and cardinality theorems.
3. Prove the dual and dual-containing theorems.
4. Prove the Gray-map isometry for the Hermitian product.
5. Run the search; tabulate results.
6. Write, then check every claim against the search output.

Risks and fallbacks.

Risk	Fallback
Non-chain ideals are too hard to classify fully	Restrict to monomial and principal ideals, and state this as the theorem's scope.
Gray map is not Hermitian-compatible	Build a twisted inner product, or use the Euclidean CSS route over F4 images instead.
Search finds no new quantum codes	Reframe as a structural paper on a new family and list near-optimal codes with defect 1 or 2.

Risk	Fallback
Result already exists elsewhere	Run a literature check on constacyclic codes over matrix rings and quantum codes before writing.

Journals to consider (check current quartiles yourself): Quantum Information Processing, Finite Fields and Their Applications, Designs Codes and Cryptography, Cryptography and Communications, Discrete Mathematics, and IEEE Transactions on Information Theory if the results are strong.